# Day of the Week Finder Worksheet 2026

Zeller steps, doomsday anchors and spreadsheet codes

| DATE | WHAT HAPPENS | DONE |
|---|---|---|
| **Step 1** | Write the date as q (day), m (month) and year. Jan. and Feb. count as months 13 and 14 of the previous year. | ☐ |
| **Step 2** | K = last two digits of that year. J = first two digits. | ☐ |
| **Step 3** | h = (q + floor(13(m+1)/5) + K + floor(K/4) + floor(J/4) + 5J) mod 7 | ☐ |
| **Step 4** | h: 0 Saturday, 1 Sunday, 2 Monday, 3 Tuesday, 4 Wednesday, 5 Thursday, 6 Friday | ☐ |
| **Check 1** | July 4, 1776: 4 + 20 + 76 + 19 + 4 + 85 = 208, and 208 mod 7 = 5, Thursday | ☐ |
| **Check 2** | Jan. 1, 2000 as 1/13/1999: 1 + 36 + 99 + 24 + 4 + 95 = 259, mod 7 = 0, Saturday | ☐ |
| **Doomsday dates** | 4/4, 6/6, 8/8, 10/10, 12/12, 5/9, 9/5, 7/11 and 11/7 share one weekday in a year | ☐ |
| **Jan. and Feb.** | Jan. 3 (Jan. 4 in a leap year) and the last day of February share it too | ☐ |
| **Century anchor** | 1800s Friday, 1900s Wednesday, 2000s Tuesday, 2100s Sunday | ☐ |
| **Year doomsday** | Anchor + floor(yy/12) + (yy mod 12) + floor((yy mod 12)/4), count forward mod 7 | ☐ |
| **2026** | Doomsday is Saturday, so July 4, 2026 is a Saturday | ☐ |
| **Spreadsheet** | Weekday of DATE(y,m,d): 1 Sunday to 7 Saturday. Type 2: 1 Monday to 7 Sunday | ☐ |
| **Julian dates** | Before Oct. 15, 1582 (Sept. 14, 1752 in Britain) swap floor(J/4) + 5J for 5 - J | ☐ |

**My notes**

Source: NIST Dictionary of Algorithms and Data Structures (Zeller's congruence); Microsoft Support and Google Docs Editors Help (weekday function); NASA and Royal Museums Greenwich (calendar changes). Formula checked by Fresh Calendars against JavaScript dates, 1583 to 2400.